「Cauchy - Riemann equation.

Let $f : G \to \mathbb{C}$ be a map on region G and let $u(x,y) = \operatorname{Re} f(x,y)$, $v(x,y) = \operatorname{Im} f(x,y)$.
Suppose that u and v have continuous partial derivatives. Then,

f is analytic if and only if u and v satisfy the Cauchy-Riemann equation.

Proof) Suppose that f is analytic. For $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+h)-f(z)}{h} = \frac{u(x+h,y)-u(x,y)}{h} + i \cdot \frac{v(x+h,y)-v(x,y)}{h}$$

Letting $h \to 0$, then $f'(z) = \frac{\partial u}{\partial x}(x,y) + i \cdot \frac{\partial v}{\partial x}(x,y)$.

Meanwhile, for $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+ih)-f(z)}{ih} = -i \cdot \frac{u(x,y+h)-u(x,y)}{h} + i \cdot \frac{v(x,y+h)-v(x,y)}{ih}$$

Letting $h \to 0$, then $f'(z) = -i \cdot \frac{\partial u}{\partial y}(x,y) + \frac{\partial v}{\partial y}(x,y)$

So, $u_x = v_y$ and $v_x = -u_y$. $\cdots (*)$ not using continuously differentiable.

Conversely, u and v satisfy the condition $(*)$, and u_x, u_y, v_x, v_y are continuous.
Let $z = x+iy \in G$ be given. Since G is open, for some $r>0$, $B_r(z) \subseteq G$.
For $h = s+it \in B_r(0)$,

$$u(x+s,y+t) - u(x,y) = [u(x+s,y+t) - u(x,y+t)] + [u(x,y+t) - u(x,y)]$$

By the Mean-Value Theorem for variables s and t, respectively,

$u(x+s,y+t) - u(x,y+t) = u_x(x+s_1, y+t) \cdot s$ for some $|s_1| < |s|$

and

$u(x,y+t) - u(x,y) = u_y(x, y+t_1) \cdot t$ for some $|t_1| < |t|$ $\quad (\star)$

Let $\varphi(s,t) = [u(x+s,y+t) - u(x,y)] - [u_x(x,y) \cdot s + u_y(x,y) \cdot t]$. Now, $(\star)$ gives

$$\frac{\varphi(s,t)}{s+it} = \frac{s}{s+it} \cdot [u_x(x+s_1, y+t) - u_x(x,y)] + \frac{t}{s+it} \cdot [u_y(x, y+t_1) - u_y(x,y)]$$

But, $|s_1| < |s| \le |s+it|$ and $|t_1| < |t| \le |s+it|$, so letting $s+it \to 0$, then

$$\frac{\varphi(s,t)}{s+it} \to 0$$

Therefore, $u(x+s,y+t) - u(x,y) = u_x(x,y)s + u_y(x,y)t + \varphi(s,t)$

and similarly $v(x+s,y+t) - v(x,y) = v_x(x,y)s + v_y(x,y)t + \psi(s,t)$ where $\lim_{s+it \to 0} \frac{\psi(s,t)}{s+it} = 0$.

Now, $$\frac{f(z+s+it)-f(z)}{s+it} = u_x(z) + i v_x(z) + \frac{\varphi(s,t) + i\psi(s,t)}{s+it} \to u_x(z) + i v_x(z)$$

as $s+it \to 0$.